AI-Based E-Commerce Product Representation Consistency and Reliability Assessment System

First Submission Document

B.Tech CSE - Major Project

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1. Abstract

Online e-commerce platforms rely on seller-provided product representations -- textual descriptions, specification bullet points, and catalog photographs -- as the primary medium through which consumers evaluate products prior to purchase. However, discrepancies between these advertised representations and actual product characteristics remain a significant source of consumer dissatisfaction and elevated product return rates. Existing platform mechanisms such as aggregate star ratings and sentiment analysis tools fail to capture claim-level inconsistencies, as they measure overall customer emotion rather than the factual accuracy of individual seller claims.

This project presents an AI-based system for assessing the consistency and reliability of e-commerce product representations by systematically cross-examining seller claims against empirical customer evidence drawn from review text and, where available, customer-uploaded photographs. Building upon the PRAISE framework (Qidwai et al., ACL 2025), which established LLM-driven attribute extraction and cross-source comparison between seller descriptions and customer reviews, the proposed system extends the paradigm by introducing a hybrid architecture combining LLM-based claim extraction with local Natural Language Inference (NLI) models for deterministic consistency scoring, lightweight computer vision techniques for visual discrepancy detection, and a unified quantitative Product Representation Index (PRI). The system categorizes each product representation into a four-tier assessment taxonomy: Consistent, Contradictory, Mixed Evidence, or Insufficient Evidence.

2. Objective & Problem Statement

2.1 Problem Statement

In contemporary e-commerce ecosystems, consumers make purchasing decisions based exclusively on digital product representations provided by sellers: textual titles, marketing descriptions, technical specification bullet points, and studio-photographed catalog images. The consumer has no physical access to the product prior to purchase and must trust that the seller's digital representation faithfully reflects the actual product.

Industry data indicates that approximately 30% of products purchased online are returned, with the primary cited reason being "item not as described" -- encompassing material misrepresentation, sizing inaccuracies, color discrepancies, and functional underperformance relative to advertised claims.

Current platform-level mechanisms are inadequate for detecting representation inconsistencies at the claim level:

Aggregate Star Ratings: Compress diverse customer experiences into a single numerical average, rendering individual claim contradictions invisible. A 5-star review stating "Beautiful product but this is polyester, not cotton" contributes a perfect score while the material lie remains hidden.

Sentiment Analysis Tools: Measure the emotional polarity of customer feedback (positive/negative) without evaluating whether specific seller claims are factually supported or contradicted by customer evidence.

Fake Review Detection Systems: Assess review authenticity (bot-generated vs. human-written) but do not evaluate whether the content of authentic reviews corroborates or refutes seller representations.

No existing deployed system systematically deconstructs seller claims into atomic propositions and cross-examines them against the full body of available customer evidence to produce claim-level consistency assessments with quantitative reliability scoring.

2.2 Objectives

1. To design and implement an automated system that extracts atomic, testable claims from e-commerce seller product descriptions and specifications using LLM-based natural language understanding.
2. To develop a hybrid text-based consistency engine that retrieves semantically relevant customer review sentences for each seller claim using Sentence-BERT and classifies the evidential relationship (Support, Contradict, or Neutral) using a pretrained Natural Language Inference (NLI) Cross-Encoder model.
3. To implement a lightweight visual consistency module using computer vision techniques (CIE LAB color distance, histogram comparison, SSIM structural similarity) to detect visual discrepancies between seller catalog images and customer-uploaded photographs, where available.
4. To aggregate claim-level textual and visual assessments into a unified, quantitative Product Representation Index (PRI: 0-100%) and assign each product a four-tier reliability status: Consistent, Contradictory, Mixed Evidence, or Insufficient Evidence.
5. To build an interactive Streamlit-based dashboard that presents claim-by-claim consistency breakdowns with supporting evidence quotes, visual comparison results, and overall PRI scores for consumer and platform use.

3. System Requirement Specification

3.1 Hardware Requirements

Component	Minimum Specification
Processor	Intel Core i3 (10th Gen) or equivalent, 2+ cores
RAM	4 GB (8 GB recommended for concurrent model loading)
Storage	10 GB free disk space (models, dataset, application)
GPU	Not required. All models run on CPU. Integrated graphics sufficient.
Network	Internet for initial model download and Gemini API access

3.2 Software Requirements

Component	Specification
Operating System	Windows 10/11, Ubuntu 20.04+, or macOS 12+
Programming Language	Python 3.10+
Deep Learning Framework	PyTorch 2.0+
NLP - Retrieval	sentence-transformers (all-MiniLM-L6-v2, 80 MB)
NLP - Inference	transformers (cross-encoder/nli-MiniLM2-L6-H768, 120 MB)
LLM API	Google Gemini 2.0 Flash (via google-generativeai SDK)
Computer Vision	OpenCV (cv2), scikit-image (SSIM), Pillow
Web Dashboard	Streamlit
Dataset	Amazon Reviews 2023 (McAuley Lab, UCSD)

3.3 Functional Requirements

ID	Requirement
FR-01	Accept seller product description (title, bullet points, specifications) as input.
FR-02	Accept a corpus of customer reviews associated with the product as input.
FR-03	Extract atomic, testable claims from seller description using Gemini Flash API.
FR-04	Retrieve top-K semantically relevant review sentences per claim using Sentence-BERT.
FR-05	Classify each claim-sentence pair as Support, Contradict, or Neutral using NLI model.
FR-06	Compute per-claim consistency scores by aggregating NLI results across all relevant sentences.
FR-07	Optionally compare catalog images with customer photos using CIE LAB, histogram, and SSIM.
FR-08	Aggregate text and visual scores into a unified PRI (0-100%).
FR-09	Assign product a 4-tier status: Consistent / Contradictory / Mixed / Insufficient Evidence.
FR-10	Display results on interactive Streamlit dashboard with evidence quotes and breakdowns.

3.4 Non-Functional Requirements

ID	Requirement
NFR-01	Process a single product (up to 200 reviews) within 120 seconds on a dual-core CPU.
NFR-02	All scoring models run locally without paid API subscriptions. Only claim extraction uses API.
NFR-03	Scoring pipeline produces deterministic, reproducible results on identical inputs.
NFR-04	System is modular: if visual engine is disabled or no photos exist, text engine functions independently.

4. Existing System

4.1 Current Industry Mechanisms and Their Limitations

a) Aggregate Star Rating Systems

Platforms like Amazon and Flipkart display an average star rating (e.g., 4.2/5.0) computed across all customer reviews. This mechanism suffers from a critical information loss problem: a 5-star review stating "Beautiful product but this is definitely polyester, not the silk claimed in the listing" contributes a perfect score to the aggregate, rendering the material contradiction invisible to prospective buyers. The single-number aggregation collapses multi-dimensional product quality attributes into one value, making it impossible to identify which specific claims are trustworthy and which are not.

b) Sentiment Analysis Tools

Academic and commercial sentiment analysis systems classify review text along an emotional polarity spectrum (Positive, Negative, Neutral). These tools measure how the customer feels about the product, not whether specific seller claims are factually accurate. A review expressing positive sentiment about design while simultaneously reporting a material or sizing discrepancy is classified as "Positive," and the factual discrepancy signal is entirely lost in the classification output.

c) Fake Review Detection Systems

Research in fake review detection (Jindal & Liu, 2008; Ott et al., 2011) focuses on identifying bot-generated or incentivized reviews. Even when all reviews in a corpus are confirmed as authentic human-written reviews, these systems provide no mechanism for evaluating whether the factual content of those authentic reviews corroborates or contradicts what the seller claims in the product listing.

d) PRAISE: LLM-Driven Attribute Comparison (ACL 2025) -- Base Paper

The most directly relevant prior work is the PRAISE system (Qidwai et al., ACL 2025) developed by researchers at IIIT Hyderabad, University of Pennsylvania, and Arizona State University. Published at ACL 2025 (the #1 ranked NLP conference globally, Core A*), PRAISE introduced a four-stage LLM pipeline:

Stage 1 - Attribute Extraction: Each customer review is processed by an LLM (Gemini 2.0 Flash) to extract objective, factual attribute-value pairs while filtering subjective opinions and irrelevant content (shipping, packaging).

Stage 2 - Cross-Comparison: Each extracted attribute is compared against the seller description and classified as Matching, Contradictory, Missing, or Partially Matching.

Stage 3 - Attribute Grouping: Semantically similar attributes are clustered under common labels (e.g., "battery lasts 8 hours", "charge dies by noon" both grouped under Battery Life).

Stage 4 - Structured Presentation: Results displayed in an organized web interface showing claim-by-claim assessments.

4.2 Limitations of PRAISE (Motivating Our System)

Limitation 1 - Text-Only Modality (Section 8): PRAISE relies exclusively on textual review content. It cannot analyze visual discrepancies between catalog imagery and customer-uploaded photographs such as color mismatches, texture differences, or design deviations.

Limitation 2 - Computational Cost (Section 3): The pipeline requires approximately 2R + 1 LLM API calls per product (where R = number of reviews). A product with 100 reviews requires 201 API calls, creating scalability and cost concerns.

Limitation 3 - No Quantitative Reliability Score: PRAISE outputs qualitative categorical tables but never produces a single numerical metric for overall product representation reliability, making quick cross-product comparison impossible.

Limitation 4 - Uniform Review Weighting: All reviews receive equal evidential weight regardless of purchase verification status, review length, or community helpfulness endorsements.

4.3 Research Gap

No existing system combines: (a) atomic claim-level deconstruction of seller descriptions, (b) semantically filtered evidence retrieval, (c) formal NLI-based consistency classification, (d) lightweight cross-modal visual verification, and (e) unified quantitative reliability scoring into a single integrated assessment platform. The proposed system addresses this gap.

5. Architecture / System Design

5.1 System Overview

The proposed system employs a hybrid, modular architecture comprising five core components: (1) LLM-based Claim Extractor, (2) Semantic Retrieval Engine, (3) NLI Consistency Classifier, (4) Visual Consistency Module, and (5) PRI Aggregator with Dashboard. Each component is independently testable and can be upgraded or replaced without affecting the rest of the pipeline.

5.2 Architectural Design Philosophy

Unlike the base paper PRAISE which uses a single LLM for all pipeline stages, our system follows a "right tool for the right job" hybrid principle: Gemini Flash API handles the linguistically complex task of understanding seller descriptions and extracting atomic claims (where LLMs excel), while lightweight local models handle the high-volume retrieval and scoring tasks (where determinism, speed, and zero-cost operation are critical). This hybrid approach reduces total API calls from 2R+1 (PRAISE) to just 1 per product, while maintaining deterministic reproducibility in the scoring pipeline.

5.3 Detailed Pipeline Architecture

Stage 1: Claim Extraction (Gemini 2.0 Flash API)

The seller's product description (title, bullet points, specification table) is sent to the Gemini 2.0 Flash API with a structured prompt instructing it to decompose the text into discrete, atomic, testable claims. Each claim is a single verifiable statement about one product attribute.

```
Input: "Men's Premium 100% Genuine Italian Leather Oxford Shoes. Non-slip rubber outsole. True to standard US sizing. Color: Rich Dark Brown. Breathable memory foam insole."

Output Claims:
  C1: "The shoe material is 100% genuine Italian leather"
  C2: "The outsole is rubber and non-slip"
  C3: "The shoe fits true to standard US sizing"
  C4: "The color is rich dark brown"
  C5: "The insole is breathable memory foam"

API Calls: 1 per product (vs. 2R+1 in PRAISE). This is the ONLY stage that uses an external API.
```

Stage 2: Semantic Retrieval (Sentence-BERT, Local)

All customer reviews are split into individual sentences. Each sentence is encoded into a 384-dimensional vector using the all-MiniLM-L6-v2 Sentence-BERT model (80 MB, runs on CPU). For each claim from Stage 1, the system computes cosine similarity between the claim vector and all review sentence vectors, retrieving the top-K (default K=20) most semantically relevant sentences.

```
For Claim C1: "The shoe material is 100% genuine Italian leather"

Retrieved review sentences (by semantic similarity):
  S1: "This is clearly plastic coated with a thin layer, not real leather" (sim=0.82)
  S2: "The leather quality is surprisingly good for this price point" (sim=0.79)
  S3: "Started peeling after 2 weeks, cheap synthetic material" (sim=0.76)
  S4: "Beautiful design, my wife loved the color" (sim=0.31) -- filtered out (below threshold)
  S5: "Fast shipping, arrived in 2 days" (sim=0.08) -- filtered out (irrelevant)
```

This retrieval step solves a critical efficiency problem: instead of feeding every review sentence to the NLI model (expensive), we only feed the top-K relevant sentences (cheap). This is a key improvement over PRAISE.

Stage 3: NLI Consistency Classification (Cross-Encoder, Local)

Each (claim, retrieved sentence) pair is fed to the cross-encoder/nli-MiniLM2-L6-H768 NLI model (120 MB, runs on CPU). This model is pretrained on 900,000+ human-annotated sentence pairs from the SNLI and MultiNLI benchmarks. It outputs three probabilities for each pair: P(Support), P(Contradict), P(Neutral).

```
Example NLI Classification for Claim C1 ("genuine Italian leather"):

Pair: C1 + "This is clearly plastic, not real leather"
  -> Support: 0.03, Contradict: 0.91, Neutral: 0.06  =>  CONTRADICT

Pair: C1 + "The leather quality is surprisingly good"
  -> Support: 0.88, Contradict: 0.05, Neutral: 0.07  =>  SUPPORT

Pair: C1 + "Started peeling after 2 weeks, cheap synthetic"
  -> Support: 0.04, Contradict: 0.87, Neutral: 0.09  =>  CONTRADICT

Claim C1 Aggregation: 1 Support, 2 Contradict, 0 Neutral
Claim C1 Score = Support / (Support + Contradict) = 1/3 = 33.3%  =>  CONTRADICTORY
```

This is the core scientific mechanism: Natural Language Inference treats each seller claim as a hypothesis and each customer review sentence as evidence, then formally tests whether the evidence supports, contradicts, or is neutral to the hypothesis. This is not sentiment analysis -- it is factual consistency verification.

Stage 4: Visual Consistency Module (OpenCV, Local, Optional)

When customer-uploaded photographs are available, the system compares them against seller catalog images using three lightweight computer vision techniques that require negligible RAM and run instantly on any CPU:

Technique	Description	RAM Usage

CIE LAB Color Distance	Converts both images to CIE LAB color space and computes Delta-E distance. Delta-E < 10: colors match. Delta-E >	< 5 MB
Histogram Comparison	Compares color distribution histograms using OpenCV's compareHist function. Detects overall color tone shifts betw	< 5 MB
SSIM (Structural Similarity)	Computes structural similarity index between images using scikit-image. SSIM = 1.0: identical. SSIM < 0.6: significan	< 5 MB

If no customer photos exist for a product, this module simply does not activate. The text engine produces the complete assessment independently. The visual module is a modular enhancement, not a dependency.

Stage 5: PRI Aggregation and 4-Tier Assessment

The Product Representation Index (PRI) aggregates all claim-level scores into a single 0-100% metric:

PRI Calculation:

Text Score = Weighted average of all individual claim scores from Stage 3
Visual Score = Average of CIE LAB, Histogram, and SSIM scores from Stage 4

If customer photos exist: PRI = 0.7 x Text Score + 0.3 x Visual Score
If no customer photos: PRI = 1.0 x Text Score

4-Tier Classification:
PRI >= 75% -> CONSISTENT (seller claims well-supported by evidence)
PRI 50% - 74% -> MIXED EVIDENCE (some claims supported, others disputed)
PRI < 50% -> CONTRADICTORY (seller claims largely refuted by evidence)
Reviews < 3 -> INSUFFICIENT EVIDENCE (not enough data to assess)

5.4 Models and Resource Summary

Model/Tool	Purpose	Size	Runs On	Cost
Gemini 2.0 Flash	Claim Extraction	Cloud API	1 call/product	Free (Google AI Studio)
all-MiniLM-L6-v2	Sentence-BERT Retrieval	80 MB	Local CPU	Free (open-source)
nli-MiniLM2-L6-H768	NLI Classification	120 MB	Local CPU	Free (open-source)
OpenCV + scikit-image	Visual Analysis	< 10 MB	Local CPU	Free (open-source)

Peak RAM Usage: Approximately 1.5 GB (with sequential model loading). The system is designed to run on hardware as modest as an Intel Core i3 with 4 GB RAM and no dedicated GPU.

5.5 Key Improvements Over Base Paper (PRAISE)

PRAISE Limitation	Our System's Solution
Text-only modality	Adds lightweight visual module (CIE LAB, Histogram, SSIM)
2R+1 API calls per product	Reduces to 1 API call; scoring is fully local
No quantitative score	Introduces PRI (0-100%) with 4-tier taxonomy
All reviews weighted equally	Architecture supports weighted scoring (future extension)
Non-deterministic LLM scoring	Deterministic NLI Cross-Encoder for scoring

6. Conclusion

The proposed AI-Based E-Commerce Product Representation Consistency and Reliability Assessment System addresses a critical gap in existing e-commerce quality mechanisms by shifting the evaluation paradigm from aggregate sentiment measurement to claim-level factual consistency assessment. By building upon the PRAISE framework (ACL 2025) and extending it with a hybrid architecture, the system achieves four key advancements: (1) reduced computational cost through semantic retrieval-based filtering and a single API call for claim extraction, (2) deterministic and reproducible scoring through local NLI models, (3) cross-modal visual verification through lightweight computer vision techniques, and (4) a unified quantitative Product Representation Index (PRI) for overall reliability assessment.

The expected outcomes of this project include: (a) a functional prototype capable of processing real-world Amazon product data from the Amazon Reviews 2023 benchmark dataset, (b) claim-by-claim consistency reports with supporting customer evidence quotes, (c) optional visual discrepancy analysis between catalog and customer photographs, (d) a unified PRI score quantifying overall representation reliability, and (e) an interactive Streamlit-based dashboard for transparent result presentation.

The system is designed to operate primarily on open-source, locally executable models (Sentence-BERT, NLI Cross-Encoder, OpenCV) with only the claim extraction stage utilizing the Gemini Flash API. This ensures accessibility on modest hardware (Intel Core i3, 4 GB RAM, no dedicated GPU), reproducibility for academic evaluation, and minimal operational cost for practical deployment.

The beneficiaries of this system include: (1) online consumers who gain evidence-backed pre-purchase trust assessments, (2) e-commerce platforms seeking to reduce return rates caused by representation inaccuracies, and (3) honest sellers who receive diagnostic feedback for improving the accuracy of their product listings.

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